Ho Chi Minh City University of Transport

Faculty: Information Technology

Department: Software Engineering

COURSE SYLLABUS

1. Course Information

Course Name	**Introduction to Information Technology**

Course Code: 122042

No of Credits	**3 (2,1,3)**										
Time Allocation	**45**	LT	**30**	BT/ TL		TN/ TH	**15**	BTL		TKMH/ DAMH	
	Outside Practice:										
Assessment Scheme (Scoring Scale: 10)	Ongoing Assessment:	**50%**	Progress tests, Assignments, Labs, Mid-term exam								
	Final Examination:	**50%**	Constructed-Response Test								
Pre-requisites	None										
Graduate Program	***Information Technology***										
Degree Level Knowledge	Bachelor Specialized										
Course policy	Students must not miss more than 20% of total course sessions										

Notes:

- *Abbreviates: LT:Lecture, BT: Assignment, TL: Discussion , TN: Experiment, TH: Lab, BTL: Capstone project, TKMH: Course design, DAMH: Course project;*
- *Capstone project: Each has no more than 1 BTL, each course has no more than 3 BTL;*
- *TKMH, DAMH: Course projects or design have own course code;*
- *Lecture: 1 credit = 15 sessions;*
- *BT,TL, TN,TH: 1 credit = 15 sessions;*

2. Course Objective:

2.1 Knowledge: This course provides essential knowledge about careers in information technologies, basic concepts of information technology and problem solving methods with computers.

2.2 Skills: Problem analysis and specification, algorithm development, implementation a program in a high-level programming language that automates solutions.

3. Course description:

This course consists of 9 chapters:

- Chapter 1: Explore possible career paths, fields, and opportunities available to a computer scientist or computer engineer. This part also introduces issues of using computer in a networked world and computer ethics.
- Chapter 2: Introduce the basic concepts of information technology, including data and information, computer system, hardware/software, network, internet and web, management data.
- Chapter 3: Introduce problem solving methods with computers.
- Chapter 4,5,6,7,8,9: Introduce basic concepts in computer programming.

4. Course Outline:

4.1 Content

Unit	Content	LT (Session)	BT/TL (Session)	TN/TH (Session)	BTL/DA (Session)	Total
I	***Part I. Introduction to IT Industry***					
1	***Chapter 1. Careers in IT***	2				2
2	***Chapter 2. Basic Concepts Of IT***	4				4
II	***Part II. Problem Solving and Programming***					
3	***Chapter 3. Problem Solving***	6				6
4	***Chapter 4. Introduction to Programming and Programming Language***	2				2
5	***Chapter 5. Elementary Programming***	4		3		7

6	***Chapter 6. Built-In Functions and Modules***	3		3		6
7	***Chapter 7. Control Flow Statements***	3		3		6
8	***Chapter 8. Defining Functions***	3		3		6
9	***Chapter 9. Strings and Lists***	3		3		6
	Total:	30		15		45

(TH: Lab; BT: Assignment; TL: Discussion; TKMH: Course Design; BTL: Capstone project; DA: Course Project)

4.2 Detailed Contents, Teaching and Assessment Methods

Knowledge	Skills	Teaching Methods	Assessment Methods
Part I. Introduction to IT Industry			
Chapter 1. Careers in IT 1.1 IT in the Organization 1.2 IT as a Discipline 1.3 Development Trends 1.4 Demand for IT Human Resources 1.5 Jobs and Skills in IT 1.6 Career Opportunities for IT 1.7 Ethics References: Page 6-10 [1] , Chapter 1,2,3[2], Chapter 9 [3]	- Gain insight into purpose of an IT department in an organization, the IT trends, possible career paths - Discuss the privacy issues and cyber crimes - Discuss computer ethics	- Presentation - Interactive Q&A - Group discussion	- Questions and answers - Quiz
Chapter 2. Basic Concepts Of IT 2.1 Data and Information 2.2 Computer Systems 2.3 Hardware 2.4 Software 2.5 Computer Network 2.6 The Internet and Web 2.7 Management of Data References: Chapter 1,2,3,4,5,6,7,8 [3],	Understand the basic concepts of IT: data and information, computer system, hardware/software, network, management of	- Presentation - Interactive Q&A - Group discussion	- Questions and answers - Exercises - Quiz

Chapter 1,2, 3,4 [4]	data		
Part II. Problem Solving and Programming			
Chapter 3. Problem Solving with Computer 3.1 Introduction to Problem Solving 3.2 Algorithm 3.3 Express an Algorithm 3.4 Basic Control Structures References: Chapter 4 [5]	- Have a familiarity with problem solving methods, including problem analysis, design, decomposition and step-wise refinement, and algorithm development.	- Presentation - Interactive Q&A - Group discussion	- Questions and answers - Exercises
Chapter 4. Introduction to Programming and Programming Language 4.1 Programs and Programming 4.2 Kinds of Errors 4.3 Generation of Programming Languages 4.4 Compiler and Interpreter 4.5 Characteristics of High-Level Language 4.6 Environments References: Chapter 13[3], Chapter 1[6]	- Understand concepts of program, programming, and programming language. - Install environments to develop program.	- Presentation - Interactive Q&A - Group discussion	- Questions and answers
Chapter 5. Elementary Programming 5.1 Keywords 5.2 Identifier 5.3 Data Types 5.4 Variables and Constants 5.5 Operators and Expressions 5.6 Statements 5.7 Comments References: Chapter 2 [6], [7]	- Write programs that perform simple computations. - Use identifiers to name elements.	- Presentation - Interactive Q&A - Group discussion	- Questions and answers - Exercises and Assignments - Quiz
Chapter 6. Built-In Functions and Modules	- Use input/output functions.	- Presentation - Interactive	- Questions and answers - Exercises

6.1 Input/Output Functions 6.2 Math Functions 6.3 Some commonly used functions 6.4 Working with External Libraries References: [7]	- Solve problems by using the functions in the libraries.	Q&A - Group discussion	and Assignments - Quiz
Chapter 7. Control Flow Statements 7.1 Control Flow 7.2 Conditional Statements 7.3 Looping Statements 7.4 "break" and "continue" Statements References: Chapter 5,9 [6], [7]	- Write programs with conditional and looping statements. - Implement program control with break and continue.	- Presentation - Interactive Q&A - Group discussion	- Questions and answers - Exercises and Assignments - Quiz
Chapter 8. Defining Functions 8.1 Organizing Program 8.2 Defining a Function 8.3 Calling a Function 8.4 Passing Arguments 8.5 Scope of Variables References: Chapter 3[6], [7]	- Design and implement functions using stepwise refinement. - Develop reusable code that is easy to read, debug, and maintain.	- Presentation - Interactive Q&A - Group discussion	- Questions and answers - Exercises and Assignments - Quiz
Chapter 9. Strings and Lists 9.1 Strings 9.2 Manipulating Strings 9.3 Lists 9.4 Manipulating Lists 9.5 List Comprehension References: Chapter 4,8[6], [7]	- Describe why lists are useful in programming - Explore common operations for sequences	- Presentation - Interactive Q&A - Group discussion	- Questions and answers - Exercises and Assignments - Quiz

4.3 Time Allocation

Content	**Time allocation for teaching and learning**					
	In class			**TH/TN**	**Home Study**	**Total**
	LT	**BT**	**TL**			
Chapter 1. Careers in IT						

1.1 IT in the Organization	0.2				0.2	0.4
1.2 IT as a Discipline	0.2				0.2	0.4
1.3 Development Trends	0.2				0.2	0.4
1.4 Demand for IT Human Resources	0.2				0.2	0.4
1.5 Jobs and Skills in IT	1				1	2
1.6 Career Opportunities for IT 1.7 Ethics	0.2				0.2	0.4
Chapter 2. Basic Concepts Of IT						
2.1 Data and Information 2.2 Computer Systems	1				1	2
2.3 Hardware 2.4 Software	1.5				1.5	3
2.5 Computer Network 2.6 The Internet and Web	1				1	2
2.7 Management of Data	0.5				0.5	1
Chapter 3. Problem Solving						
3.1 Introduction to Problem Solving 3.2 Algorithm	0.5				0.5	1
3.3 Express an Algorithm	0.5				0.5	1
3.4 Basic Control Structures	5				5	10
Chapter 4. Introduction to Programming and Programming Language						
4.1 Programs and Programming 4.2 Kinds of Errors	0.5				0.5	1
4.3 Generation of Programming Languages 4.4 Compiler and Interpreter	0.5				0.5	1
4.5 Characteristics of High-Level Language	0.5				0.5	1

4.6 Environments	0.5				0.5	1
Chapter 5. Elementary Programming						
5.1 Keywords 5.2 Identifier	0.5			0.5	1	2
5.3 Data Types 5.4 Variables and Constants	1			1	2	4
5.5 Operators and Expressions	2			1.5	3.5	7
5.6 Statements 5.7 Comments	0.5				0.5	1
Chapter 6. Built-In Functions and Modules						
6.1 Input/Output Functions	1			1	2	4
6.2 Math Functions	1			1	2	4
6.3 Some commonly used functions 6.4 Working with External Libraries	1			1	2	4
Chapter 7. Control Flow Statements						
7.1 Control Flow	0.5					0.5
7.2 Conditional Statements	1			1	2	4
7.3 Looping Statements	1			1	2	4
7.4 “break” and “continue” Statements	0.5			1	1.5	3
Chapter 8. Defining Functions						
8.1 Organizing Program	0.5					0.5
8.2 Defining a Function	1			1	2	4
8.3 Calling a Function	1			1	2	4

8.4 Passing Arguments						
8.5 Scope of Variables	0.5			1	1.5	3
Chapter 9. Strings and Lists						
9.1 Strings 9.2 Manipulating Strings	1.5			1.5	3	6
9.3 Lists 9.4 Manipulating Lists	1			1	2	4
9.5 List Comprehension	0.5			0.5	1	2

5. Course Materials

[1]. Christine Wilcox – "Careers in Information Technology", ReferencePoint Press, Inc., 2015.

[2]. Adriano Fabris – "Ethics of Information and Communication Technologies", Springer, 2018.

[3]. Timothy J. O'Leary, Linda I. O'Leary, Daniel A. O'Leary – "Computing Essentials", McGraw-Hill Education, 2017

[4]. Reema Thareja, "Fundamentals of Computers", Oxford University Press, 2016

[5]. Misty E. Vermaat, "Discovering computers 2016, Tool, Apps, Devices, and the Impact of Technology", Cengage Learning, 2015.

[6]. Paul Gries, Jennifer Campbell, Jason Montojo, "Practical Programming: An Introduction to Computer Science Using Python 3.6 (3rd Edition)", Pragmatic Bookshelf, 2017

[7]. https://docs.python.org/3/tutorial/

6. Learning Outcomes

On successful completion of this course, students will be able to:

- Explain the purpose of an IT department in an organization
- Gain insight into the IT trends, possible career paths and the skills needed for different types of jobs.
- Discuss the privacy issues and cyber crimes including creation of malicious programs as well as denial of service attacks, internet scams, identity theft.
- Discuss computer ethics including copyright law, software piracy, digital rights management,…
- Differentiate between information and data

- Understand the functions and roles of operating systems in the management of computer processes and data.
- Describe the different types of computer hardware and software.
- Explain the basic concepts of the network, internet and web.
- Understand Management of Data.
- Have a familiarity with problem solving methods, including problem analysis, design, decomposition and step-wise refinement, and algorithm development.
- Explain the fundamental concepts in programming, including data types and data structures, control flow, repetition, selection, input/output, and functions.
- Write programs in a high-level programming language that automates solutions.

7. Assessment Scheme

- Ongoing Assessment: 50%, including attendance point, progress tests, assignments, labs, mid-term exam.
- Final Examination: 50%, constructed-response test.

8. List of trainers

- Lecturers:

  Tran Thi My Tien

  Bui Trong Hieu

  Le Van Quoc Anh
- Teaching Assistants:
- Lab Instructors:

Ho Chi Minh City, October 23, 2022

DEAN OF THE FACULTY	HEAD OF THE DEPARTMENT	SYLLABUS CREATOR
Le Van Quoc Anh	**Le Van Quoc Anh**	**Tran Thi My Tien**